

Module 11: Genomics and Biotechnology - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. What does PCR do?
2. A. cuts DNA into random pieces
3. B. reads the sequence of a protein
4. C. makes many copies of a selected DNA sequence
5. D. measures the pH of a sample
6. Answer: C

7. Why: PCR amplifies a chosen DNA region so there is enough material to analyze.

8. Gel electrophoresis separates DNA fragments by:
9. A. size, with smaller fragments moving farther
10. B. color of the fragments
11. C. temperature alone
12. D. how much protein they contain
13. Answer: A

14. Why: An electric field pulls DNA through the gel. Smaller fragments move faster and travel farther.

15. CRISPR is best described as a tool that can:
16. A. only measure the size of a genome
17. B. create energy for cells
18. C. replace the whole field of biology
19. D. target and edit specific DNA sequences
20. Answer: D

21. Why: CRISPR, adapted from a bacterial defense system, lets scientists edit DNA at chosen locations.

22. Why does biotechnology require bioethics?

23. A. because the tools never actually work

24. B. because powerful genetic tools raise questions of benefit, harm, and rights

25. C. because DNA is not real

26. D. because ethics stops all research

27. Answer: B

28. Why: Bioethics weighs benefits, risks, and responsibilities so that powerful genetic tools are used wisely.